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Program : BSSE (A)

Task 05

1. Design subnets.

An organization has:

192.168.50.0/24

It needs:

Department	Hosts Required
Data Science	40
Administration	18
Faculty	28
Management	13

Determine a suitable prefix and subnet mask for each. Then assign address ranges.

Department	Prefix	Mask	Network	Host Range	Broadcast
Data Science	/26	255.255.255.192	192.168.50.0	192.168.50.1 – 192.168.50.62	192.168.50.63
Administration	/27	255.255.255.224	192.168.50.96	192.168.50.97 – 192.168.50.126	192.168.50.127
Faculty	/27	255.255.255.224	192.168.50.64	192.168.50.65 – 192.168.50.94	192.168.50.95
Management	/28	255.255.255.240	192.168.50.128	192.168.50.129 – 192.168.50.142	192.168.50.143

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2. Subnetting Challenge

You are given:

192.168.10.0/24

The organization requires:

- Network A → 100 hosts
- Network B → 50 hosts
- Network C → 20 hosts
- Network D → 10 hosts

Design an appropriate addressing plan. Students must determine:

- Prefix
- Subnet mask
- Network address
- First usable host
- Last usable host
- Broadcast address

Network	Prefix	Mask	Network Address	First Usable	Last Usable	Broadcast
A	/25	255.255.255.128	192.168.10.0	192.168.10.1	192.168.10.126	192.168.10.127
B	/26	255.255.255.192	192.168.10.128	192.168.10.129	192.168.10.190	192.168.10.191
C	/27	255.255.255.224	192.168.10.192	192.168.10.193	192.168.10.222	192.168.10.223
D	/28	255.255.255.240	192.168.10.224	192.168.10.225	192.168.10.238	192.168.10.239